HCI 271: CAPSTONE I

User Psychology for Design

Understanding the Human Behind the Interface

Based on Seif El-Nasr, Carstensdottir & John (2026), Part II
& Don Norman (2004), Emotional Design

Reflections

Norman’s Book

Design of Everday things, Emotional Design

SESSION ROADMAP

Why Psychology Is a Design Tool

01

Cognition & Perception

How users see, process, and remember information.

02

Emotional Design

Visceral, behavioral, and reflective levels of experience.

03

Motivation (SDT)

Autonomy, competence, and relatedness as drivers.

04

Adoption (TAM)

Why users accept or reject new technology.

> ***"Design without psychology is just decoration. Understanding the human mind transforms design from aesthetic to strategic."***

COGNITIVE FOUNDATIONS

What the Brain Can (and Cannot) Do

Working Memory

The bottleneck of cognition. Can only hold **4±1 chunks** of information at once. Overloading it causes errors and abandonment.

Long-Term Memory

Stores mental models and schemas. Design should leverage existing models (e.g., "folder" icon) to enable **recognition over recall**.

Attention

Selective and limited. Users do not read; they scan. Visual hierarchy, contrast, and motion must **direct attention** to key tasks.

Cognitive Load Theory

Intrinsic: Task difficulty.
Extraneous: Bad design (distractions).
Germane: Learning effort.
Goal: **Minimize extraneous load.**

Design Implication: Don't make users think. Use chunking, progressive disclosure, and consistent patterns to respect cognitive limits.

PERCEPTUAL ORGANIZATION

The Brain Groups Before It Reads

Proximity

Close elements form a group.

Similarity

Like elements are grouped visually.

Continuity

Align elements to guide the eye.

Closure

Mind fills missing information.

Layout, whitespace, and grouping communicate structure before words.

Source: Designing Serious Games; Wertheimer (1923)

User Psychology

Discussion

What color combinations you should not use in your designs?

DON NORMAN'S FRAMEWORK

Emotion Is Not a Feature — It Is the Experience

1. Visceral

"Does it look good?"

Pre-conscious & Automatic. First impressions, aesthetics, and immediate emotional reaction.

2. Behavioral

"Does it work well?"

Use & Experience. Pleasure of effectiveness: function, performance, and usability.

3. Reflective

"What does it mean?"

Conscious & Meaning. Self-image, satisfaction, and memories that build loyalty.

"Attractive things work better." Positive affect broadens attention and increases tolerance for minor usability issues.

Source: Norman (2004), Emotional Design

User Psychology

Discussion

Why is emotional design important?

EMOTIONAL DESIGN APPLIED

Designing Across All Three Levels

VISCERAL

"The Look"

Immediate sensory impact. Colors, shapes, typography. Creates the first impression within milliseconds.

> **Example:** Apple's unboxing experience. The friction of the box, the white space, the smell.

BEHAVIORAL

"The Feel"

Usability and performance. Does it do what I want? Feedback loops, error prevention, speed.

> **Example:** Google Search. Instant results as you type. Pure function and efficiency.

REFLECTIVE

"The Meaning"

Identity and memory. What does this product say about me? Nostalgia, pride, brand loyalty.

> **Example:** Spotify Wrapped. Users share it because it tells a story about *them*.

⚠️ **Key Principle: Positive affect broadens attention (good for creativity). Negative affect narrows attention (good for focused, critical tasks).**

Source: Norman (2004), Emotional Design

MOTIVATION THEORY

Motivation Is the Engine of Engagement

Intrinsic Motivation

Doing an activity for its inherent satisfaction. The user engages because it is fun, interesting, or challenging.

Curiosity & Exploration
Mastery & Growth

Extrinsic Motivation

Doing an activity to attain a separable outcome. The user engages to earn a reward or avoid punishment.

Points, Badges, Leaderboards
Money or Grades

THE SELF-DETERMINATION CONTINUUM

Amotivation	External	Introjected	Identified	Integrated	Intrinsic

Non-Self-Determined — Self-Determined

The Overjustification Effect

Adding expected extrinsic rewards (like money or points) to an already intrinsically motivating task can **reduce** intrinsic motivation. When the reward stops, the behavior stops.

Source: Designing Serious Games, Ch. 5.5; Deci & Ryan (1985)

SELF-DETERMINATION THEORY (SDT)

Three Needs That Drive Motivation

Autonomy

"I chose to do this."

Feeling in control of one's actions; acting with **volition** in line with personal values.

Competence

"I am capable."

Feeling effective and experiencing mastery through optimal challenge and clear feedback.

Relatedness

"I belong here."

Feeling connected to others; seen in community, social proof, or contributing to a larger cause.

Research Insight: Ryan, Rigby & Przybylski (2006) show these needs independently predict enjoyment and future engagement.

Source: Designing Serious Games, Ch. 5.5; Ryan & Deci (2000)

APPLIED PSYCHOLOGY

Designing for Autonomy, Competence, Relatedness

Autonomy

DESIGN STRATEGY

Provide meaningful choices, customization options, and non-linear navigation paths. Avoid forced tutorials.

UI EXAMPLES

- "Skip Intro" button
- Avatar customization
- Dark/Light mode toggle

Competence

DESIGN STRATEGY

Offer clear feedback, visualize progress, and scaffold complex tasks. Ensure the interface is forgiving (Undo).

UI EXAMPLES

- Progress bars
- "Undo" toast notification
- Success animations

Relatedness

DESIGN STRATEGY

Facilitate social connection, cooperation, and recognition. Show that real people are behind the system.

UI EXAMPLES

- User profiles & bios
- Collaborative editing
- "X people viewed this"

Source: Designing Serious Games, Ch. 5.5

TECHNOLOGY ACCEPTANCE MODEL (TAM)

Users Adopt Technology When It Feels Useful

Perceived Ease of Use

"Is it free of effort?"

→

Perceived Usefulness

"Does it improve my performance?"

→

Actual System Use

Adoption & Engagement

Perceived Usefulness (PU)

The degree to which a person believes that using a particular system would enhance their job performance. **This is the strongest predictor of adoption.**

Perceived Ease of Use (PEOU)

The degree to which a person believes that using a particular system would be free of effort. **Ease of use enables usefulness.**

"Users will tolerate a difficult interface if the utility is high enough (e.g., Excel), but they will abandon a useless interface no matter how beautiful it is."

Source: Designing Serious Games, Ch. 5.3; Davis (1989)

DIAGNOSING ABANDONMENT

TAM Explains Why Users Leave

🚫 Failure Mode: "The Wall"

THE PROBLEM (LOW EASE OF USE)

User encounters a login screen, complex form, or paywall **before** seeing any value.

↓

THE FIX (LAZY REGISTRATION)

Allow users to use the core feature first. Ask for signup only when they want to save their work.

🚫 Failure Mode: "The Ghost Town"

THE PROBLEM (LOW USEFULNESS)

User lands on a dashboard with no data ("Cold Start"). It looks empty and useless.

↓

THE FIX (PRE-POPULATION)

Provide sample data, templates, or a "Get Started" wizard to demonstrate value immediately.

> ***"Users don't quit because it's hard; they quit because the effort isn't worth the perceived value."***

INDIVIDUAL DIFFERENCES

The "Big Five" Personality Traits

Openness

Preference for novelty, variety, and creativity.

→ **Design for discovery, customization, and non-linear exploration.**

Conscientiousness

Preference for order, duty, and achievement.

→ **Design for efficiency, clear progress tracking, and goal completion.**

Extraversion

Preference for social stimulation and energy.

→ **Design for social sharing, leaderboards, and community interaction.**

Agreeableness

Preference for cooperation and social harmony.

→ **Design for collaboration, gifting, and supportive feedback loops.**

Neuroticism

Tendency to experience negative emotions (anxiety).

→ **Design for safety, error prevention, and reassuring confirmation.**

Source: Designing Serious Games, Ch. 5.4; Costa & McCrae (1992)

SOCIAL PSYCHOLOGY

Humans Are Social: Design Should Reflect That

Social Presence

The sense of being with others, even in a digital space. It reduces isolation and increases trust.

UI PATTERNS

"User is typing..." indicators, live viewer counts, avatars on active documents.

Social Proof

The psychological phenomenon where people copy the actions of others to undertake behavior in a given situation.

UI PATTERNS

Reviews, testimonials, "Best Seller" badges, "People also bought..."

Cooperation

Designing for interdependence. Shared goals create stronger bonds than competition alone.

UI PATTERNS

Team missions, collaborative workspaces, shared playlists, community challenges.

Insight: Social motivation (Relatedness) often overrides individual efficiency. Users will endure bad UI to be with their friends.

Source: Designing Serious Games, Ch. 5.5; Cialdini (1984)

SYNTHESIS

A Framework for Psychologically-Informed Design

Cognition *"Can I process this?"*

REDUCE LOAD

Chunk content, use clear hierarchy and familiar models to avoid overwhelm.

Emotion *"Do I like this?"*

VISCERAL & BEHAVIORAL

Aesthetics should build trust; interactions should avoid frustration.

Motivation *"Do I want this?"*

SDT SUPPORT

Offer meaningful choices, timely feedback, and social connection.

Adoption *"Is it worth it?"*

TAM ALIGNMENT

Show clear utility and reduce friction before request for commitment.

Source: Synthesis of Course Concepts

User Psychology

FINAL THOUGHT

Design Is Communication

"Design is really an act of communication, which means having a deep understanding of the person with whom you are communicating."

Don Norman

THE DESIGN OF EVERYDAY THINGS